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#include <iostream>
#include <iomanip>
#include <vector>
#include <string>
using namespace std;
double balance = 0;
struct Transaction {
   int id;
   string type;
   double amount;
   string description;
};
vector<Transaction> transactions;
int nextId = 1;
void displayHeader() {
   cout <<
"\n@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@\n";
   cout << "@@        MIZAN TEPI UNIVERSITY        @@\n";
   cout << "@@            TEPI CAMPUS             @@\n";
   cout << "@@  SCHOOL OF COMPUTING & INFORMATICS @@\n";
   cout << "@@ DEPARTMENT OF INFORMATION TECHNOLOGY @@\n";
   cout << "@@         ETHIO TELECOM SYSTEM        @@\n";
   cout << "@@——————————@@\n";
   cout << "@@ Name & ID                          @@\n";
   cout << "@@ 1. Estifanos Getawey  8057/17     @@\n";
   cout << "@@ 2. Getachew Birhan    8018/17     @@\n";
   cout << "@@ 3. Abreham Tadess     7962/17     @@\n";
   cout << "@@ 4. Abule Dulo         8968/17     @@\n";
   cout << "@@ 5. Asaye Birhan       8279/17     @@\n";
   cout <<
"@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
\n\n";
}
int binarySearch(int key) {
   int low = 0, high = transactions.size() - 1;

   while (low <= high) {
      int mid = (low + high) / 2;

      if (transactions[mid].id == key)
         return mid;
      else if (transactions[mid].id < key)
         low = mid + 1;
      else
         high = mid - 1;
   }
   return -1;
```

```
}
void mainMenu() {
    displayHeader();
    cout << "1. Check Balance\n";
    cout << "2. Buy Package\n";
    cout << "3. Recharge\n";
    cout << "4. View Transactions\n";
    cout << "5. Search\n";
    cout << "6. Update\n";
    cout << "7. Delete\n";
    cout << "8. Exit\n";
    cout << "Choice: ";
    }
void recharge(int amount) {
    displayHeader();

    if (amount <= 0) {
        cout << "Invalid amount!\n";
        return;
    }

    balance += amount;
    transactions.push_back({nextId++, "Recharge", (double)amount, "Top-up"});

    cout << "Recharge Successful!\n";
    cout << "Balance: " << balance << " ETB\n";
}
void buyPackage(double price, string desc) {
    displayHeader();

    if (price > balance) {
        cout << "Insufficient balance!\n";
        return;
    }
    balance -= price;
    transactions.push_back({nextId++, "Purchase", price, desc});

    cout << "Package Purchased!\n";
    cout << "Balance: " << balance << " ETB\n";
}
void displayTransactions() {
    displayHeader();

    if (transactions.empty()) {
        cout << "No transactions found!\n";
        return;
    }
    for (auto t : transactions) {
        cout << "ID: " << t.id
            << " | " << t.type
            << " | " << t.amount
            << " | " << t.description << endl;
    }
}
void searchTransaction() {
    displayHeader();

    int id;
    cout << "Enter ID: ";
    cin >> id;

    int index = binarySearch(id);

    if (index != -1) {
        auto t = transactions[index];
        cout << "Found!\n";
        cout << t.id << " | " << t.type << " | " << t.amount << " | " << t.description <<
endl;
    } else {
        cout << "Not Found!\n";
    }
}
void updateTransaction() {
    displayHeader();

    int id;
    cout << "Enter ID: ";
    cin >> id;

    int index = binarySearch(id);

    if (index != -1) {
        cin.ignore();
        cout << "New Description: ";
        getline(cin, transactions[index].description);
        cout << "Updated!\n";
    } else {
        cout << "Not Found!\n";
    }
}
void deleteTransaction() {
    displayHeader();

    int id;
    cout << "Enter ID: ";
    cin >> id;

    int index = binarySearch(id);

    if (index != -1) {
        transactions.erase(transactions.begin() + index);
        cout << "Deleted!\n";
    } else {
        cout << "Not Found!\n";
    }
}
void packageMenu() {
    int choice;

    do {
        displayHeader();
        cout << "1. Internet\n2. Voice\n3. SMS\n4. Back\n";
        cin >> choice;

        if (choice == 1) {
            cout << "1. 1GB(15)\n2. 5GB(50)\n3. 10GB(75)\n";
            cin >> choice;
            if (choice == 1) buyPackage(15, "1GB");
            else if (choice == 2) buyPackage(50, "5GB");
            else if (choice == 3) buyPackage(75, "10GB");
        }

    } while (choice != 4);
}
int main() {

    int choice;

    do {
        mainMenu();
        cin >> choice;

        switch (choice) {
            case 1:
                displayHeader();
                cout << "Balance: " << balance << " ETB\n";
                break;

            case 2:
                packageMenu();
                break;

            case 3: {
                int amount;
                cout << "Enter amount: ";
                cin >> amount;
                recharge(amount);
                break;
            }

            case 4:
                displayTransactions();
                break;

            case 5:
                searchTransaction();
                break;

            case 6:
                updateTransaction();
                break;

            case 7:
                deleteTransaction();
                break;

            case 8:
                cout << "Thank you!\n";
                break;

            default:
                cout << "Invalid choice!\n";
        }

    } while (choice != 8);

    return 0;
}
```